

Artificial Intelligence

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What Is Artificial Intelligence?

▣ Artificial Intelligence, Machine Learning, and Deep Learning

- We need to clearly define AI, machine learning, and deep learning, and explain how they are related.

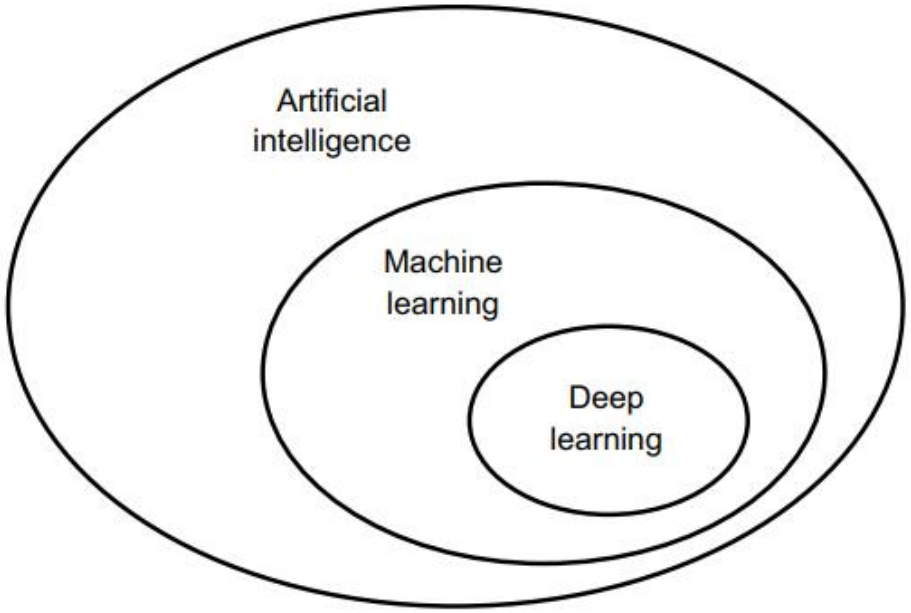


Figure 1.1 Artificial intelligence, machine learning, and deep learning

1.1.1 Artificial Intelligence

▣ Definition & Origin

- Born in the 1950s, formalized at the 1956 Dartmouth Workshop.
- AI = automating intellectual tasks normally performed by humans.

▣ Approaches

- **Symbolic AI** (1950s–1980s): rule-based systems, peaked with Expert Systems.
- Limitation: struggled with complex, fuzzy tasks (e.g., vision, speech, language).

▣ Significance

- Foundation for shift to Machine Learning (ML) and Deep Learning (DL), which drive today's AI.

1.1.2 Machine Learning

▣ Historical Roots

- Ada Lovelace (1843): observed that machines can only do what they are told—cannot create original ideas.
- Alan Turing (1950): introduced the Turing Test and raised questions about machine intelligence.

▣ From Rules to Learning

- **Classical programming:** humans write explicit rules → computer executes.
- **Machine learning:** computer is trained with data + answers → learns rules automatically.
- Paradigm shift: system generalizes from examples instead of relying on hand-coded rules.

1.1.2 Machine Learning

▣ Machine Learning

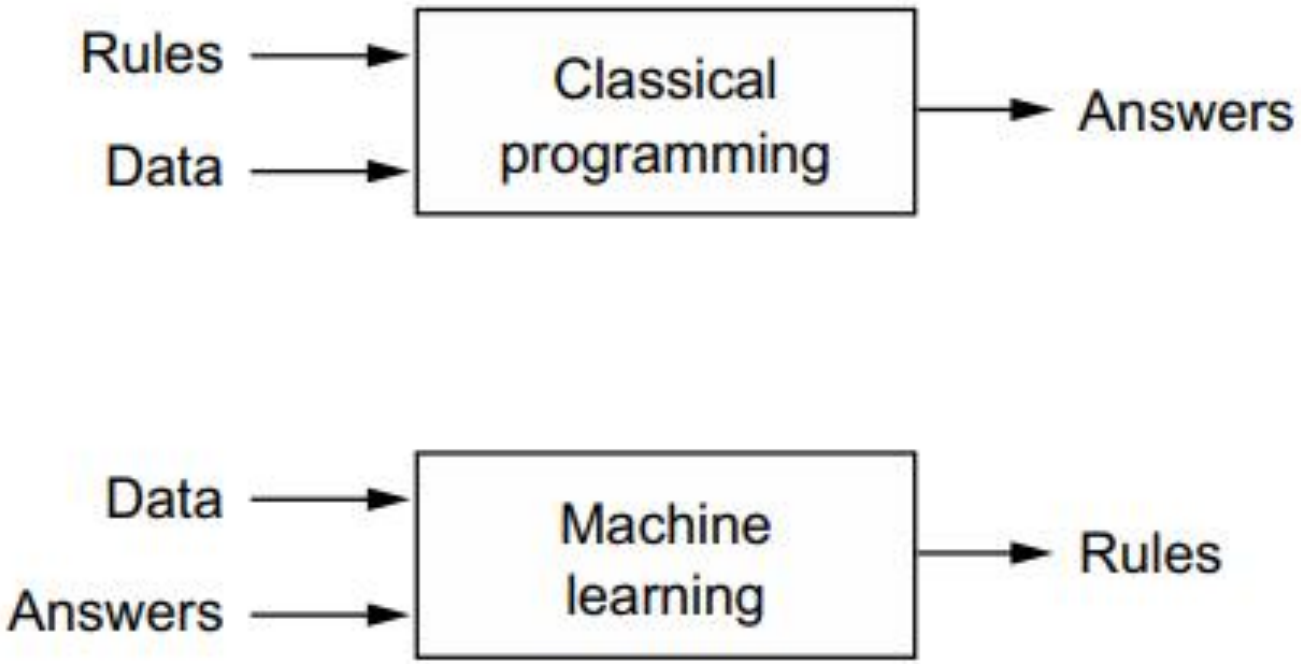


Figure 1.2 Machine learning: a new programming paradigm

1.1.2 Machine Learning

▣ Rise of ML

- Flourished in the 1990s with faster hardware & larger datasets.
- Became the most successful subfield of AI.

▣ Key Features

- Related to statistics, but more engineering-oriented.
- Handles large, complex datasets (e.g., millions of images).
- Uses statistical learning instead of handcrafted rules.
- Deep learning (subset of ML) → especially effective for complex tasks.

▣ Significance

- Shifted AI from symbolic reasoning to data-driven approaches.
- Core driver of today's AI revolution.

1.1.3 Learning Representations from Data

▣ Learning Representations from Data

- To define deep learning and understand how it differs from other ML approaches, first consider what ML algorithms do.
- Given samples and target values, ML seeks operational rules for a data-processing task. To perform ML you need three things:
  - ① **Input data points:** e.g., for **speech recognition**, the data points are **audio files** of recorded conversations; for **image tagging**, the data points are **photographs**
  - ② **Expected outputs:** for speech recognition, **human-produced transcripts** of the audio files; for image tasks, tags such as **"dog," "cat,"** etc.
  - ③ **A way to measure performance:** to determine the difference between the algorithm's current output and the expected output.

1.1.3 Learning Representations from Data

▣ Learning Representations from Data

- Let's look at a concrete example. Consider data points plotted in an **(x, y)** coordinate system as shown in **Figure 1-3**.
- As shown, there are **white** points and **red** points. We want an algorithm that takes a point's coordinates **(x, y)** as input and outputs whether the point is **red** or **white**. Summarized:
  - **Inputs:** point coordinates
  - **Expected outputs:** the point's **color**
  - **Performance measure:** the **proportion** of points correctly classifie

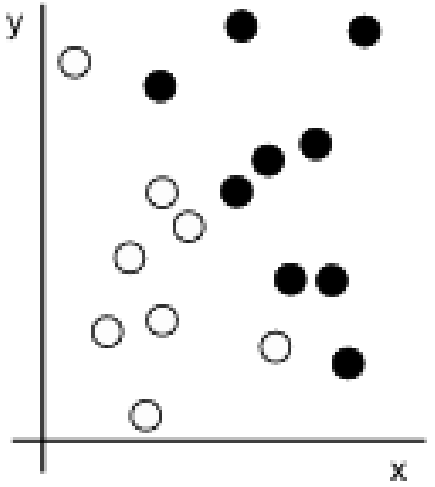


Figure 1.3 Some sample data

1.1.3 Learning Representations from Data

▣ Learning Representations from Data

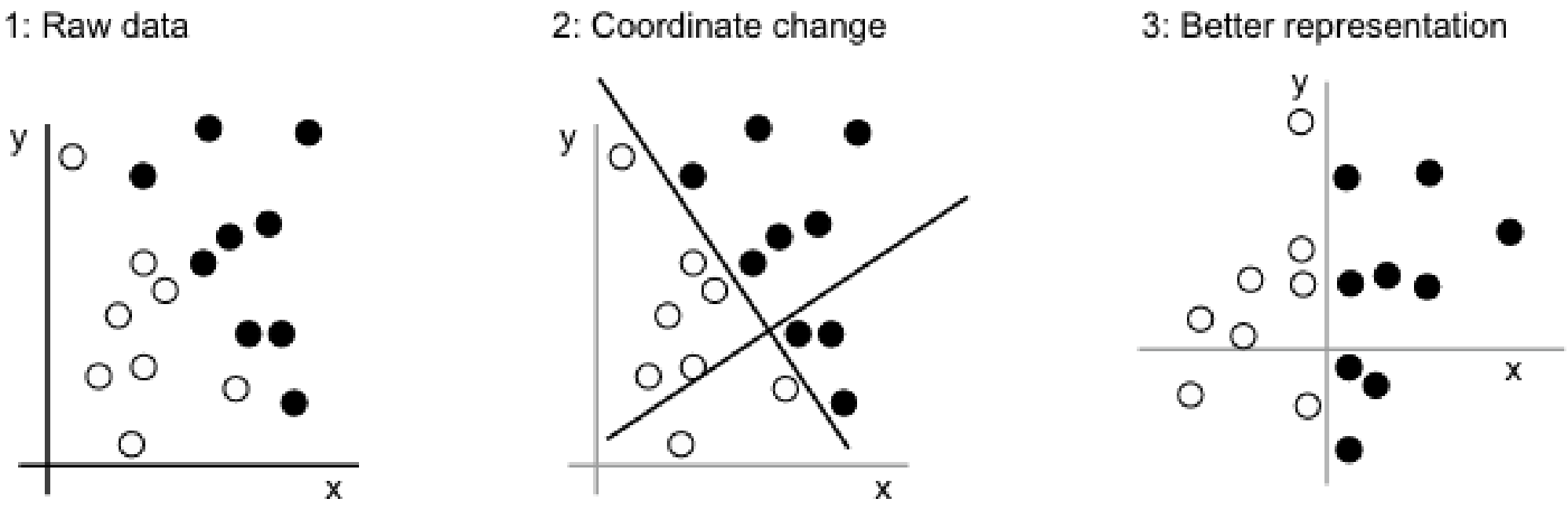


Figure 1.4 Coordinate change

1.1.3 Learning Representations from Data

▣ Learning Representations from Data

- In this case **we** specified the coordinate transform.
- A human devised an appropriate representation for the data.
- That's fine for a very simple problem, but could we do the same for **handwritten-digit classification**?
- Could we write explicit, machine-executable image transforms that capture the differences between **6** and **8**, or between **1** and **7**, across all kinds of handwriting?

1.1.4 The “deep” in “deep learning”

▣ What is “Deep” in Deep Learning?

- Deep learning is a type of machine learning.
- Focus: learn many layers of representations from data
- “Deep” = many layers stacked together (depth of model)
- Other names:
  - ➢ Layered representation learning
  - ➢ Hierarchical representation learning
- Inspired by the brain, but not the same as the brain
- Important: deep learning = math framework, not biology

1.1.4 The “deep” in “deep learning”

▣ What is “Deep” in Deep Learning?

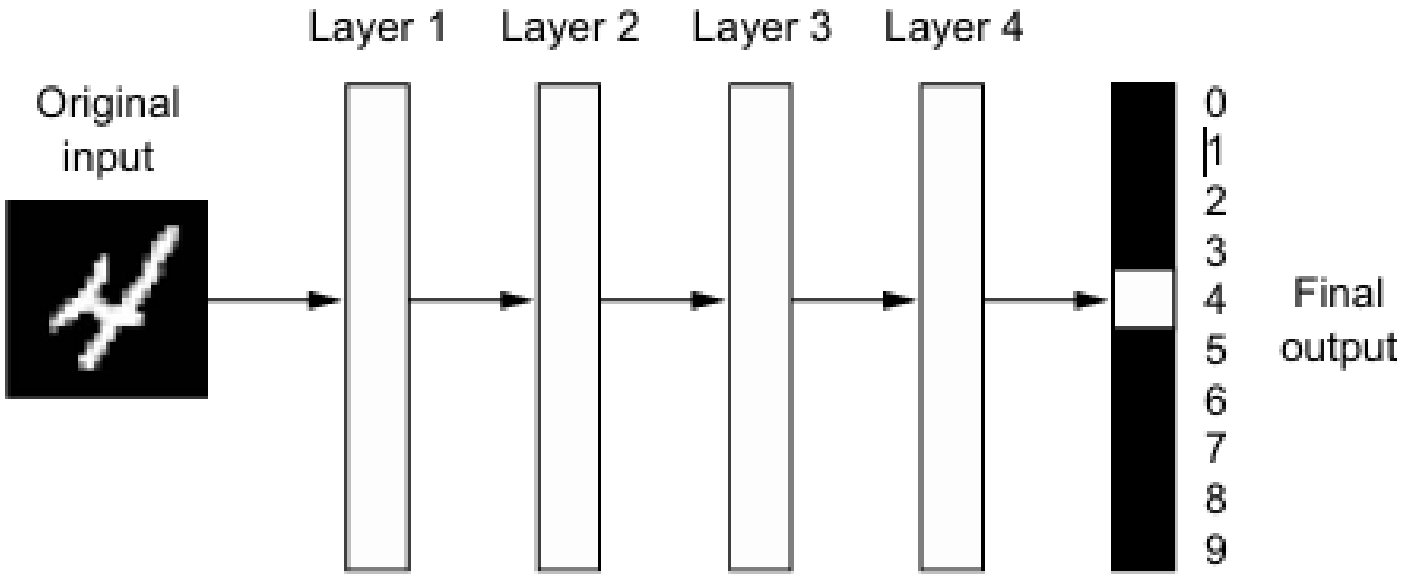
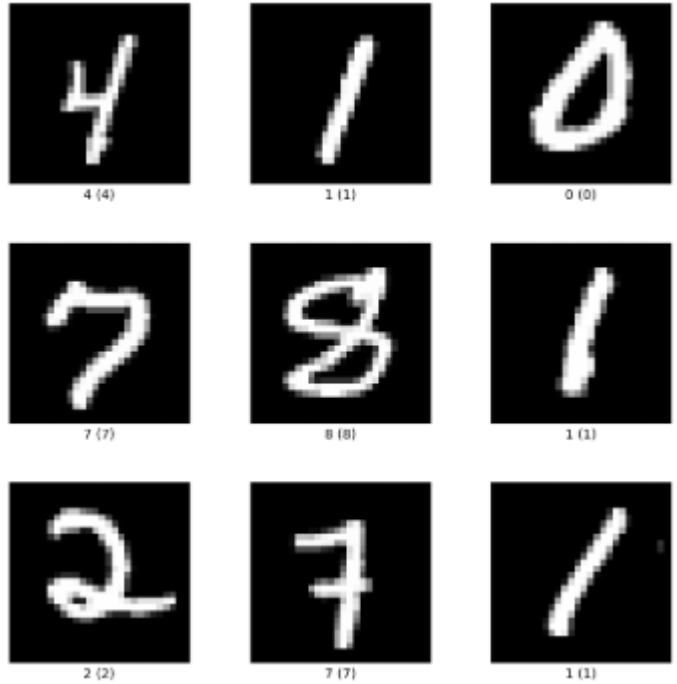


Figure 1.5 A deep neural network for digit classification

1.1.4 The “deep” in “deep learning”

How Deep Learning Learns

- Deep learning = multi-stage process
- Each layer changes the input → more useful for the task
- Example: digit recognition (MNIST)
  - Layer 1: captures edges and simple shapes
  - Layer 2: finds bigger patterns (parts of digits)
  - Layer 3: combines into full digit structure
  - Layer 4: outputs final classification (0–9)
- Think of it as information distillation:
- Data → goes through filters → purified into useful form
- Simple steps, when stacked many times, look like magic



1.1.4 The “deep” in “deep learning”

▣ How Deep Learning Learns

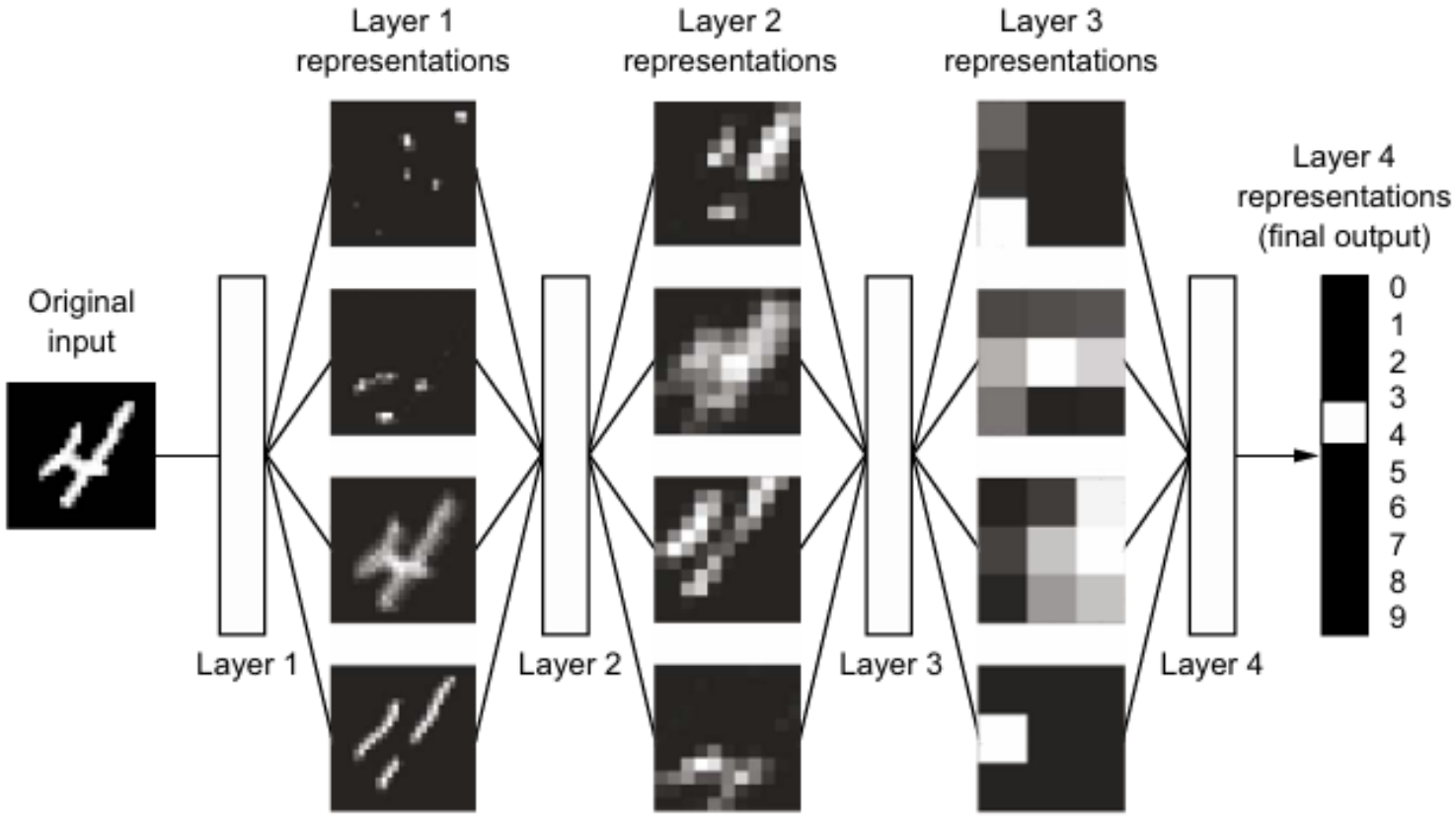


Figure 1.6 Data representations learned by a digit-classification model

1.1.5 Understanding how Deep learning works, in three figures

▣ How Deep Learning Learns (Weights & Parameters)

- Machine learning = mapping inputs (e.g., images) to targets (e.g., label "cat")
- Deep learning does this through many layers of data transformations
- Each layer's behavior is controlled by weights
  - Weights = numbers that define how data is transformed
  - A layer is **parameterized** by its weights (parameters = weights)
- Learning = finding the right weight values
- Correct weights → correct mapping from input to target
- Networks can have millions of weights
- Changing one weight affects others → makes training difficult

1.1.5 Understanding how Deep learning works, in three figures

▣ How Deep Learning Learns (Weights & Parameters)

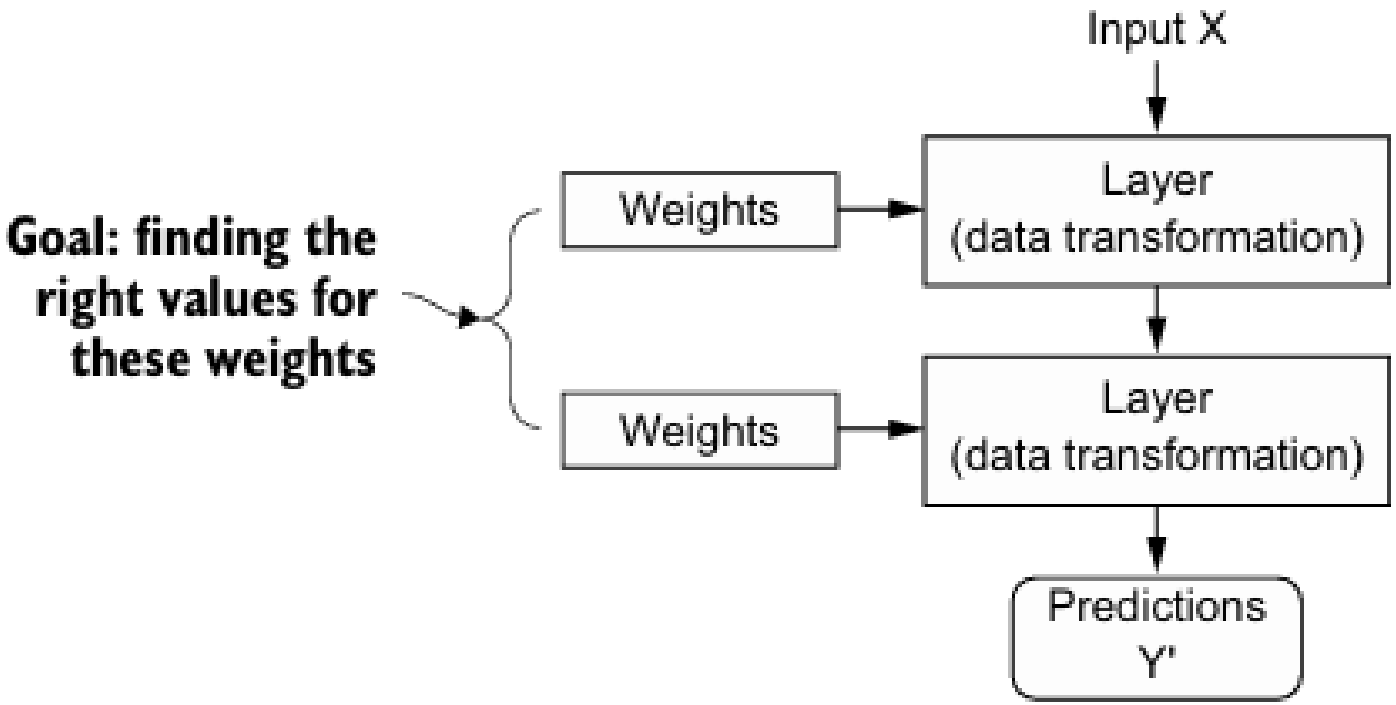


Figure 1.7 A neural network is parameterized by its weights.

1.1.5 Understanding how Deep learning works, in three figures

▣ How Deep Learning Learns (Weights & Parameters)

- To control a system, you must be able to observe it
- In neural networks:
  - ➢ We need to measure how far predictions are from the true answer
- **Loss function** = tool to measure this difference
  - ➢ Also called objective function or cost function
- How it works:
  - ➢ Takes prediction (output of the network)
  - ➢ Compares with true target (what we expected)
  - ➢ Returns a distance score → shows how well the network performed
- This score guides the learning process

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▣ How Deep Learning Learns (Weights & Parameters)

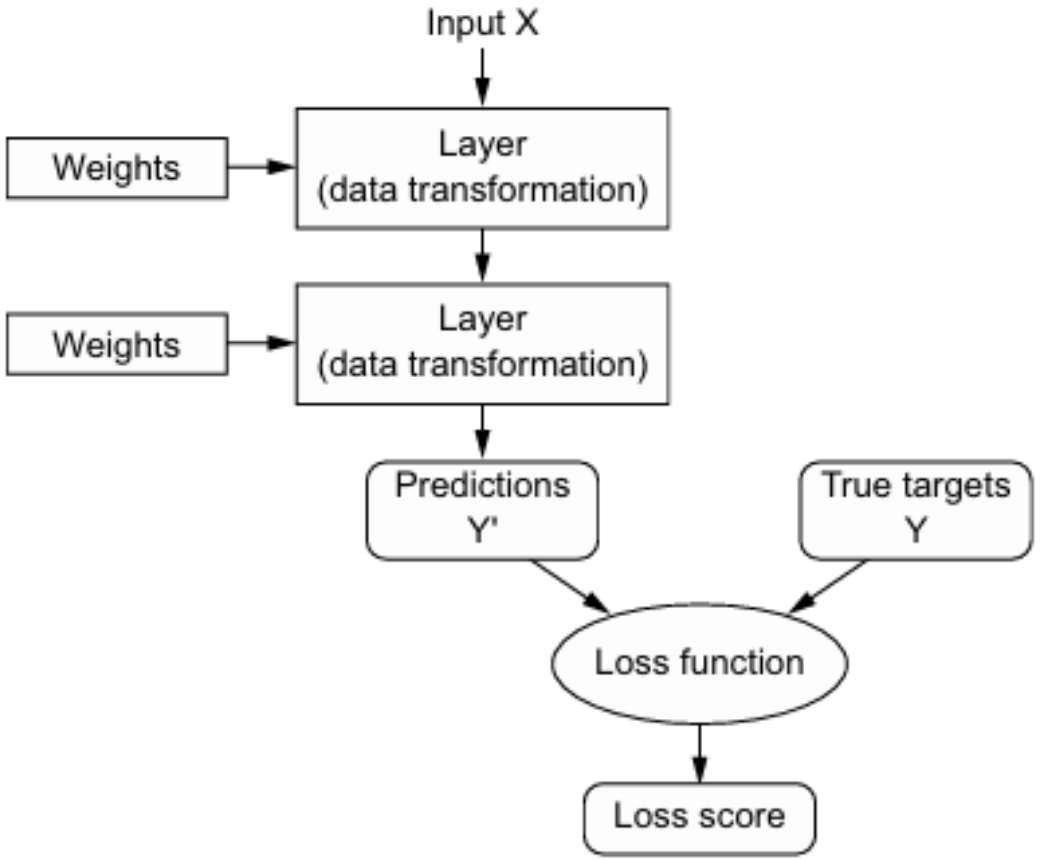


Figure 1.8 A loss function measures the quality of the network's output.

1.1.5 Understanding how Deep learning works, in three figures

▣ How Deep Learning Learns (Weights & Parameters)

- Key idea in deep learning:
  - ➢ Use the loss score as feedback
- Optimizer's job:
  - ➢ Adjust the weights slightly
  - ➢ Goal = make the loss smaller
- This process is repeated many times → network learns better predictions
- Method used: **Backpropagation**
  - ➢ The core algorithm of deep learning
  - ➢ Calculates how each weight should change

1.1.5 Understanding how Deep learning works, in three figures

▣ How Deep Learning Learns (Weights & Parameters)

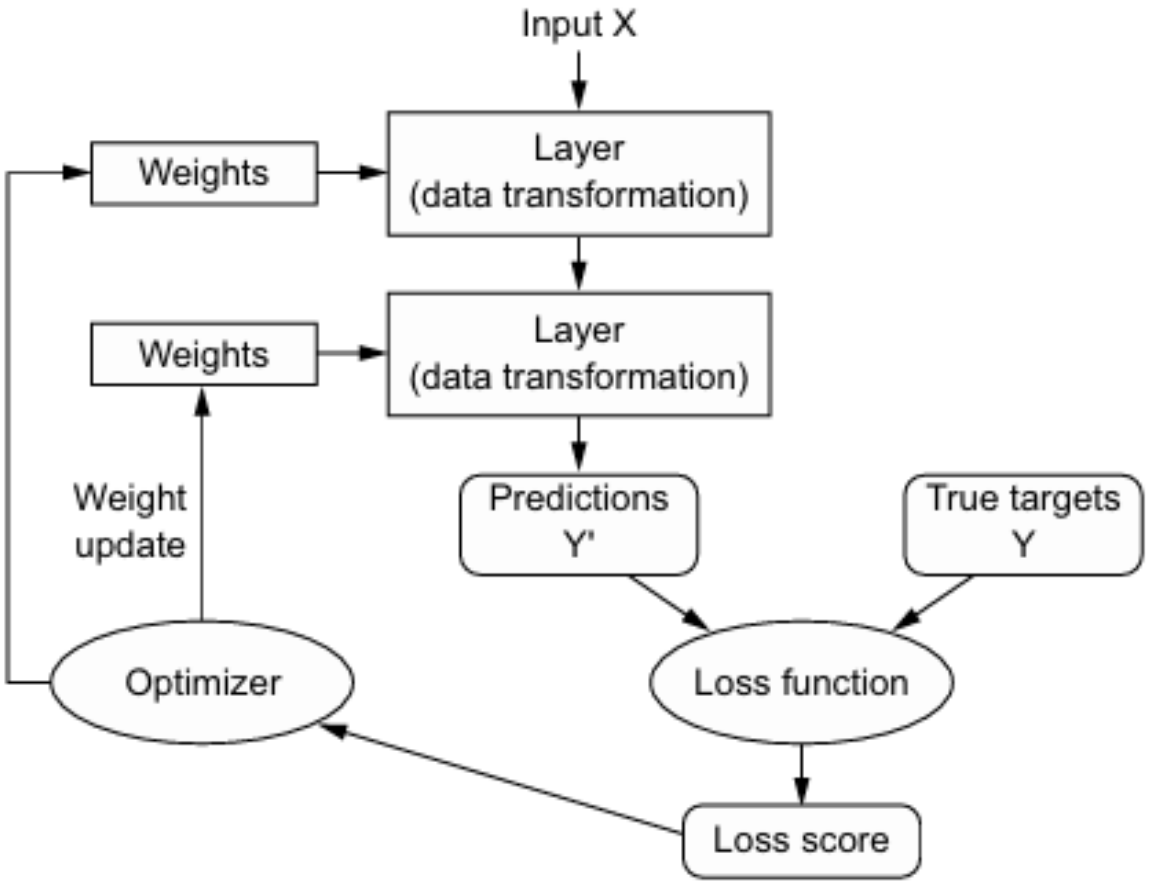


Figure 1.9 The loss score is used as a feedback signal to adjust the weights.

1.1.5 Understanding how Deep learning works, in three figures

▣ How Deep Learning Learns (Weights & Parameters)

① **At the start:**

- ✓ Weights = random values
- ✓ Network output = random / wrong
- ✓ Loss score is very high

② **During training:**

- ✓ For each example → weights adjusted a little
- ✓ Loss score gets smaller step by step

③ **After many iterations:**

- ✓ Weights reach values that minimize loss
- ✓ Network outputs ≈ true targets
- ✓ This is a trained network

④ **Key idea:**

- ✓ A simple repetitive process
- ✓ When scaled, it looks like magic

1.1.6 What deep learning has achieved so far

▣ Became widely used since early 2010s

▣ Achieved a revolution in AI:

- Tasks once hard for machines but easy for human

▣ Key Breakthroughs

- **Near-human-level:**
  - ➢ Image classification
  - ➢ Speech transcription
  - ➢ Handwriting transcription

1.1.6 What deep learning has achieved so far

▣ Other Achievements

- Much better machine translation
- Improved text-to-speech systems
- Digital assistants (Google Assistant, Alexa)
- Near-human-level autonomous driving
- Improved ad targeting & search results
- Answering natural language questions
- Beating humans in Go playing

1.1.6 What deep learning has achieved so far

▣ Applications in the real world

- Transcribing ancient manuscripts
- Diagnosing diseases via medical images
- Detecting plant diseases with smartphones
- Predicting natural disasters (floods, earthquakes, hurricanes)
- Supporting science, medicine, energy, transport, agriculture, art, and more

1.1.7 Don’t believe the short-term hype

- **Deep learning → big progress, but expectations often too high**
- **Some goals still far away:**
  - Believable dialogue systems
  - Human-level translation & language understanding
  - Human-level general intelligence (AGI)
- **History of AI hype & disappointment**
  - 1960s–70s: Symbolic AI → high hopes, then 1st AI winter
  - 1980s–90s: Expert systems → investment, then 2nd AI winter
- **Today:**
  - Possible 3rd cycle of hype
  - Need to moderate expectations
  - Focus on realistic progress, not exaggerated promises

1.1.8 The promise of AI

▣ Short-term:

- AI progress is fast, but still limited in daily life
- Current tools = useful but only **accessories** (e.g., voice assistants, search, recommendations)

▣ Long-term vision:

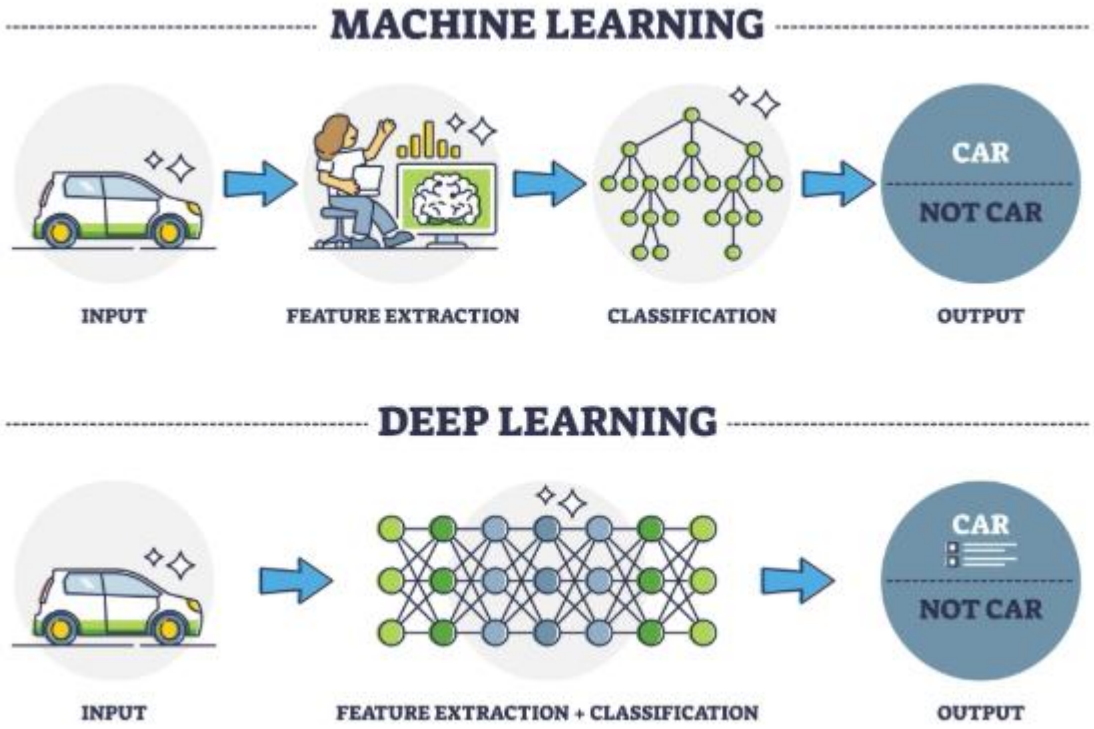
- AI will become **central** to how we work, think, and live
- Future AI will:
  - ➢ Be your assistant and helper
  - ➢ Answer questions, teach kids, watch health
  - ➢ Deliver goods, drive cars, manage complex tasks
- Support scientists → **new discoveries** in all fields (medicine, genomics, math, etc.)

▣ Key message:

- Don’t believe the **short-term hype**
- But do believe in the **long-term impact** → AI will transform society like the Internet did

1.2.1 What Makes Deep Learning Different

- **Better performance than older methods**
- **Simplifies problem-solving by automating feature engineering**
  - Shallow learning → required humans to design features
  - Deep learning → learns all features automatically



1.2.1 What Makes Deep Learning Different

▣ Key advantages:

- Joint feature learning
  - ➢ All layers learn together, not step by step
  - ➢ Features adapt automatically with one feedback signal
- Layer-by-layer representations
  - ➢ Builds complex features gradually
  - ➢ Each layer refines the data a bit more

▣ Result:

- **End-to-end learning with one model**
- More powerful & scalable than stacking shallow models

1.2.2 The modern machine learning landscape

▣ The modern machine learning landscape

- Kaggle competitions provide a good view of what machine learning methods work best in practice.
- These contests are highly competitive, with thousands of participants and big prizes.
- Surveys show that top teams usually rely on deep learning (Keras) or gradient boosted trees (LightGBM, XGBoost).

1.2.2 The modern machine learning landscape

▣ The modern machine learning landscape

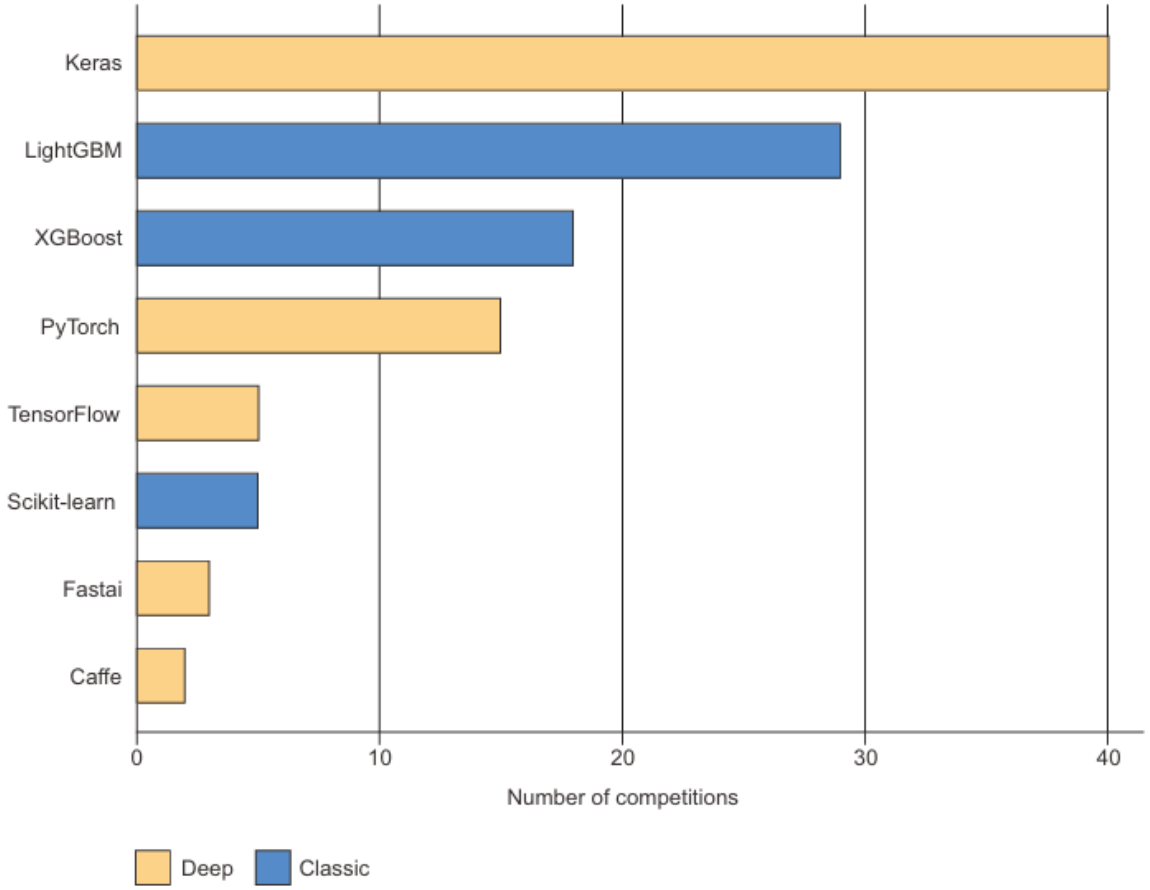


Figure 1.12 Machine learning tools used by top teams on Kaggle

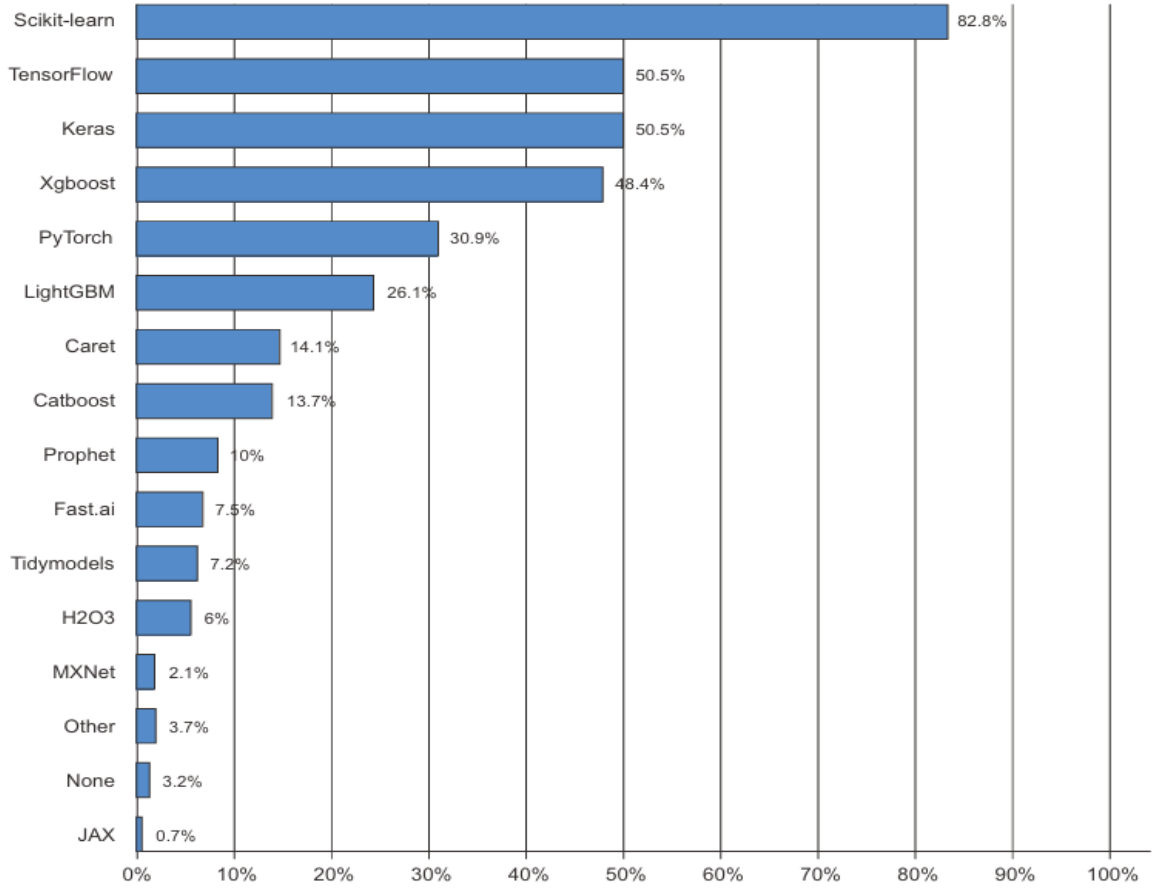


Figure 1.13 Tool usage across the machine learning and data science industry (Source: www.kaggle.com/kaggle-survey-2020)

1.3 Why deep learning? Why now?

▣ Why deep learning? Why now?

- Key ideas (CNNs, backpropagation, LSTM) existed before 2000s
- But deep learning only took off after 2012
- Three main drivers of progress:
  - **Hardware** (high-performance GPUs from gaming industry)
  - **Datasets & benchmarks** (large-scale data became available)
  - **Algorithmic advances** (new methods + scaling old ones)
- ML progress depends on experiments + resources, not just theory
- Bottlenecks of 1990s–2000s = data & hardware → solved in 2010s

1.3.1 A new wave of investment

▣ Before 2011:

- AI investment < $1 billion
- Focused mostly on shallow ML methods

▣ After deep learning breakthrough (2012–2013):

- Investments skyrocketed
- 2015: ~$5 billion
- 2017: ~$16 billion
- Hundreds of AI startups launched

1.3.1 A new wave of investment

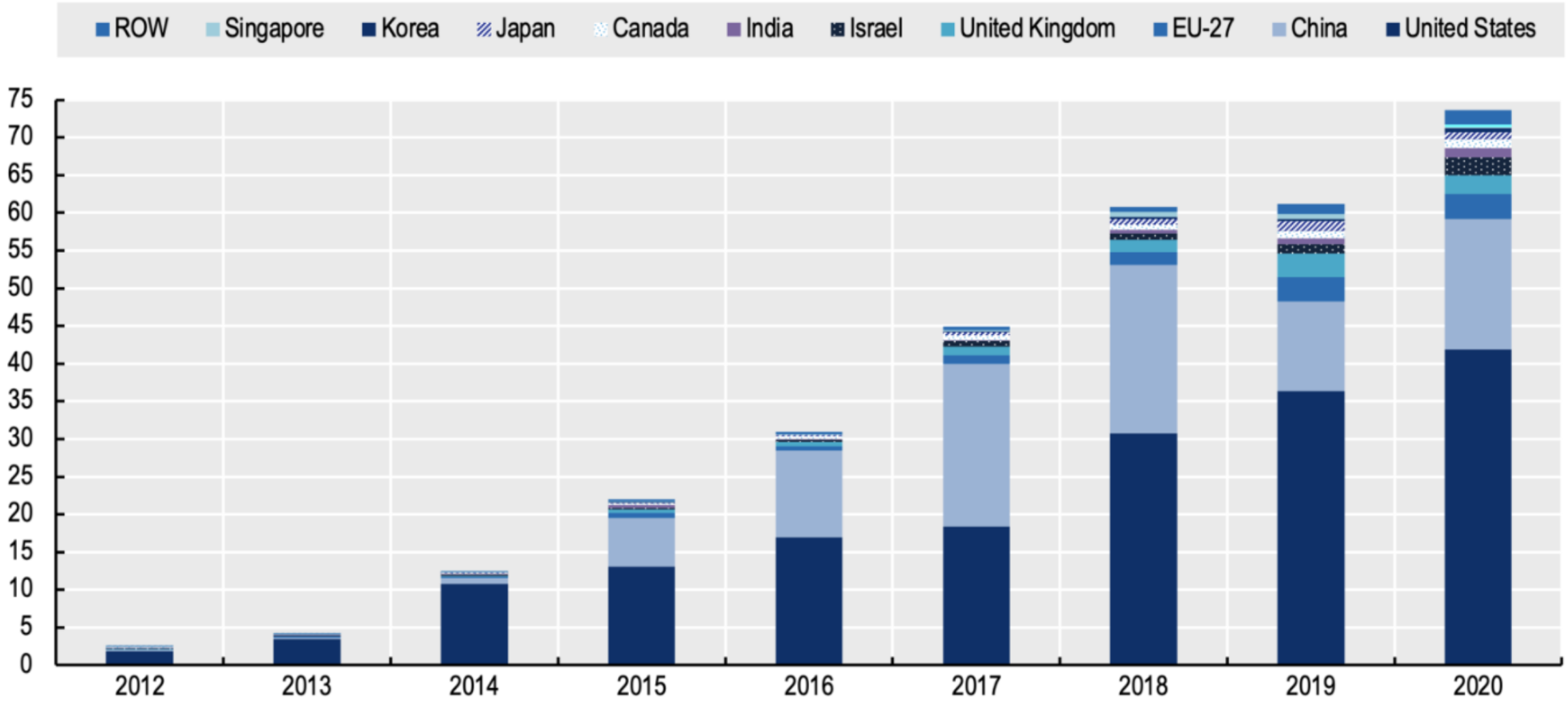
▣ Big tech companies (Google, Amazon, Microsoft):

- Invested even more through internal R&D
- AI became central to product strategy
- “Machine learning is a core, transformative way...” – Sundar Pichai (2015)

▣ Impact:

- Rapid growth in AI research & applications
- Number of researchers grew from hundreds → tens of thousands in <10 years

1.3.1 A new wave of investment



Source: OECD.AI (2021), processed by JSI AI Lab of Slovenia, based on Preqin data of 23/04/2021, www.oecd.ai.

1.3.2 The democratization of deep learning

Early days:

- Needed C++ & CUDA expertise → limited to few experts

Now:

- Basic Python skills enough for research
- Libraries simplified deep learning:
  - Theano → early symbolic tensor library
  - TensorFlow → supported autodifferentiation, easier model building
  - Keras (since 2015) → very user-friendly, like LEGO bricks

Impact:

- Lowered entry barrier → more startups, students, researchers joined
- Keras became the go-to tool for deep learning beginners and professionals

1.3.3 Will it last?

▣ Will Deep Learning Last?

- Key question: Is deep learning just hype, or a long-term revolution?
- Core strengths:
  - ➢ Simplicity → removes manual feature engineering, enables simple end-to-end models
  - ➢ Scalability → works well with GPUs/TPUs, can handle massive datasets
  - ➢ Versatility & Reusability → trained models can be reused, fine-tuned, or adapted for new tasks
- Result: Supports continuous learning and powerful production-scale systems

1.3.3 Will it last?

The Future of Deep Learning

- Deep learning still early in its journey
- Progress often follows a sigmoid curve:
  - Fast initial growth → slower incremental improvements later

Recent breakthroughs:

- Transformers in NLP (major revolution)
- Steady progress in vision & speech recognition

Outlook:

- Significant progress still ahead
- Deep learning = still a revolution in the making
- Full potential will take many years to realize



Q & A

Prof. Chang-Hee Han